

Mitchell Wortsman, Gabriel Ilharco, Samir Ya Gadre, Rebecca Roelofs, Raphael Gontijo-Lopes, Ari S Morcos, Hongseok Namkoong, Ali Farhadi, Yair Carmon, Simon Kornblith, et al. 2022. Model soups: averaging weights of multiple fine-tuned models improves accuracy without increasing inference time. In *International Conference on Machine Learning*, pages 23965–23998. PMLR.

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Michael JQ Zhang and Eunsol Choi. 2023. Mitigating temporal misalignment by discarding outdated facts. *arXiv preprint arXiv:2305.14824*.

A Appendix

A.1 Yearly Misalignment with Other Tasks and T5 Sizes

In this section, we report raw performance degradation over time on four downstream and three language modeling tasks with three sizes of T5. We evaluate on all tasks in the main paper plus Newsroom Source Classification (NewsCls) and AI Venue Classification (AIC) from [Luu et al. \(2022\)](#). We also create a third science domain language modeling task from abstracts in the Kaggle arXiv dataset⁴. For each group of three years from 2006-2008 to 2018-2020 we randomly sample 26-38M and 2.6-3.9M BPE tokens (150MB and 15MB of text) of arXiv paper abstracts for train and test splits respectively.

Figures 8 and 11 are yearly degradation heatmaps for each model size and task. These results show that normalizing performance by the average on each evaluation time helps account for variations in test splits. ArXiv language modeling and NewsSum, for example, have large differences in performance on evaluation years regardless of finetuning year.

A.2 Task Variations in Linear Yearly Degradation

Like [Luu et al. \(2022\)](#), we find differences across domain and task in the rate and linearity of year-to-year decay. TD scores measure the average rate of performance degradation for each year of misalignment between train and test time periods ([Luu et al., 2022](#)). We find the rate of decay using a linear least

squares regression and average rates for each task over all evaluations. Table 4 shows TD scores ([Luu et al., 2022](#)) for all tasks and T5-sizes. We also compare TD scores calculated from raw performance to TD scores calculated from performance normalized by the average on each evaluation year. In general, percent performance difference from the mean on an evaluation year decays more linearly than raw performance.

A.3 Yearly and Monthly Cosine Similarities

In this section, we report cosine similarity between each pair of yearly and monthly time vectors. Figure 10 shows cosine similarity between every pair of year vectors for each T5-size and task. Figure 9 shows cosine similarity between each pair of T5-small monthly WMT LM time vectors. Similar to performance, year-to-year degradation in cosine similarity between task vectors appears to be linear regardless of setting. Like figure 3, we observe seasonal "stripes" every 12 months from the diagonal 9.

A.4 Temporal Degradation in Online Settings

Our work so far illustrates temporal misalignment on static time splits. However, in practice, we usually deploy language models in online settings, meaning that they are continually updated with the latest data, and we do not have access to data from all training years simultaneously.

To show how temporal misalignment manifests in these settings, we first sort all the training data from the PoliAff and WMT tasks by month, and finetune T5-small on each task separately. We display the performance of the LM on every year throughout training in Figure 13. As expected, for PoliAff, we see that the performance of models on a particular year peak at the final month of that year, and then gradually degrade as the model continues training.

For language modeling on WMT data, performance consistently *improves* during training, regardless of the evaluation year. However, perplexity reduces more slowly in earlier years as we continue training. These results suggest that temporal misalignment may manifest differently in online settings based on the training setup and task.

A.5 Online Cosine Similarities

We study the relationship between performance degradation and cosine similarity during *online*

⁴<https://www.kaggle.com/datasets/Cornell-University/arxiv/data>

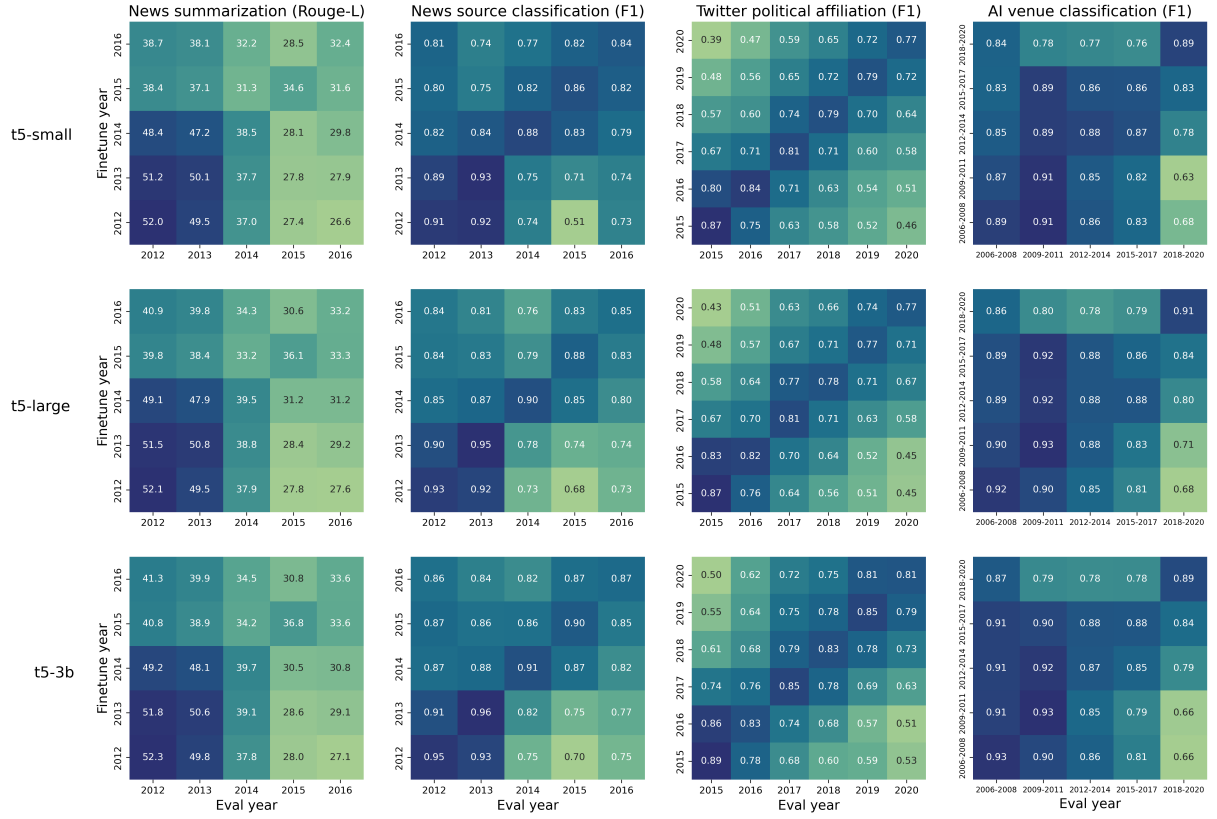


Figure 8: Yearly downstream performance degradation on four tasks and three T5 sizes.

Normalized?	T5 Size	WMT LM	NewsSum	NewsCls	Twitter LM	PoliAff	ArXiv LM	AIC
No	small	-0.67 (0.81)	2.21 (0.51)	0.05 (0.67)	-0.35 (0.97)	0.08 (0.98)	-0.59 (0.65)	0.03 (0.55)
	large	-0.10 (0.34)	2.07 (0.53)	0.04 (0.61)	-0.20 (0.97)	0.07 (0.97)	-0.20 (0.67)	0.03 (0.50)
	3b	-0.07 (0.34)	2.12 (0.53)	0.04 (0.67)	-0.20 (0.97)	0.07 (0.95)	-0.13 (0.66)	0.03 (0.40)
Yes	small	-1.70 (0.90)	6.99 (0.87)	6.43 (0.74)	-4.52 (0.89)	10.47 (0.95)	-2.61 (0.94)	2.93 (0.57)
	large	-0.56 (0.92)	6.27 (0.89)	5.33 (0.84)	-2.64 (0.91)	9.57 (0.94)	-1.24 (0.93)	2.53 (0.51)
	3b	-0.52 (0.93)	6.44 (0.88)	4.72 (0.84)	-2.90 (0.91)	7.66 (0.91)	-0.96 (0.94)	3.12 (0.61)

Table 4: TD scores for all tasks and T5 sizes for raw performance and performance divide by the average on each eval. year. Variance explained by the TD score linear fit in parentheses. TD scores calculated with normalized performance decay have generally higher R^2 scores, except on Twitter LM and PoliAff, and are easier to compare.

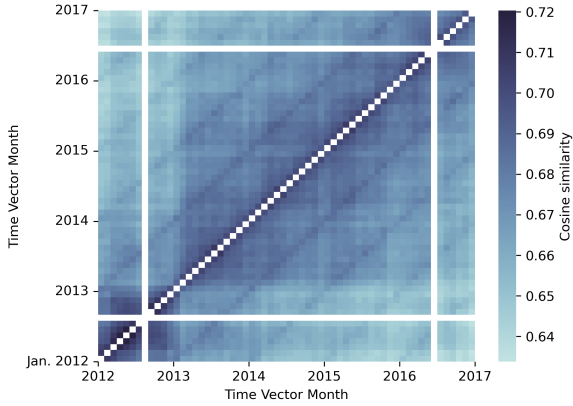


Figure 9: **Cosine similarity between monthly time vectors also exhibits seasonality.** We observe similar "stripes" every 12 months when measuring the cosine similarity between each pair of T5-small WMT month vectors. The correlation between this heatmap (including the diagonal) and figure 3 is -0.667 with $p < 1 \times 10^{-16}$.

training. Recall that in the online setting, we perform a single finetuning run on the PoliAff and WMT tasks (after ordering their training data by month), and measure performance on each year throughout training. To study how time vectors move throughout space in this setting, we measure the cosine similarity between the time vector of the model trained up to month m and each yearly time vector for the PoliAff and WMT tasks.

We find that the cosine similarity to each time vector *decreases* as the online model is updated past the first 12 months of data. This means that online models' peak similarity to earlier years tends to be higher than those to later years since they make up a smaller part of its total finetuning set. Like our experiments with soups of time vectors in section §4.5, this indicates that models trained on multiple years of data lay outside a region defined by single-year models.

To account for these decreases, we normalize the similarity to each year time vector by its average after updating on all months in Figure 13. Our results reveal that the vector for our online model is relatively most similar to each year vector after finetuning on the months in that year.

A.6 Time Vector Analogy Ablations

In this section, we ablate our time vector analogy experiment to determine the effects of only adding the LM vector from the target time, and only scaling the weights of the initial time vector. For $\tau_k \approx \alpha_1 \cdot \tau_j + (\alpha_2 \cdot \tau_k^{\text{LM}} - \alpha_3 \cdot \tau_j^{\text{LM}})$, we define our "task

addition" ablation for $\alpha_3 = 0, \alpha_1, \alpha_2 \neq 0$, and our "scaling only" ablation for $\alpha_1 \neq 0, \alpha_2, \alpha_3 = 0$

We report the best results after sweeping over the same α ranges from §4.4 with the added constraints in figure 15. While task analogies generally perform best across tasks and T5-sizes (especially as τ_j and τ_k become more misaligned), we find that ablating τ_k^{LM} and τ_j^{LM} can still improve over the base τ_j model. Surprisingly, *only scaling* τ_j also improves over the initial model on many tasks.

A.7 Temporal Misalignment Affects Some Parameters More than Others

In this section, we explore whether we can reduce temporal misalignment by swapping parameter weights from a model trained on a misaligned year with those of the model trained on the target year. For example, we substitute the QKV attention layers from a model finetuned on 2015 PoliAff with those finetuned on 2020 PoliAff and evaluate on 2020 data. In table 5 we evaluate the start-year finetuned models for each task on the end times (e.g. start = 2012 for WMT LM, end = 2016) with various parameter weights swapped with the end-year finetuned model.

From these experiments, we find that we can improve performance on a target time by swapping out weights with a time vector finetuned on that time. Surprisingly, swapping embeddings with the target time vector makes very little difference, except in language modeling tasks, and swapping all non-embedding weights with a target time almost reaches the performance the target time-specific models for downstream tasks. Swapping only feed-forward or attention layers also improves performance on the target time, suggesting temporal misalignment is somewhat isolated to those model regions in downstream tasks.